1) a)

$A\ \left(2, \frac{\pi}{6}\right)$

$B\ \left(4, \frac{5\pi}{12}\right)$

$C\ \left(8, \frac{3\pi}{4}\right)$

$D\ \left(3, -\frac{2\pi}{3}\right)$

$E\ \left(5, -\frac{\pi}{2}\right)$

$F\ \left(\frac{5}{2}, -\frac{\pi}{6}\right)$

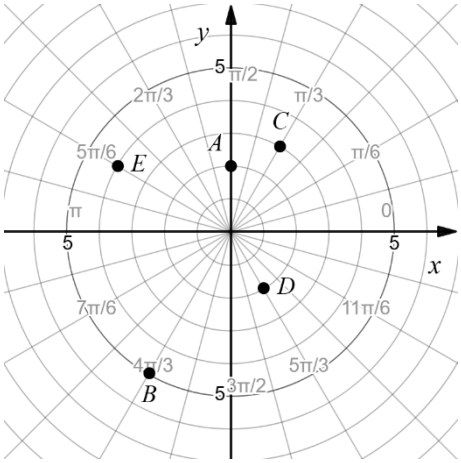
1) b)

C

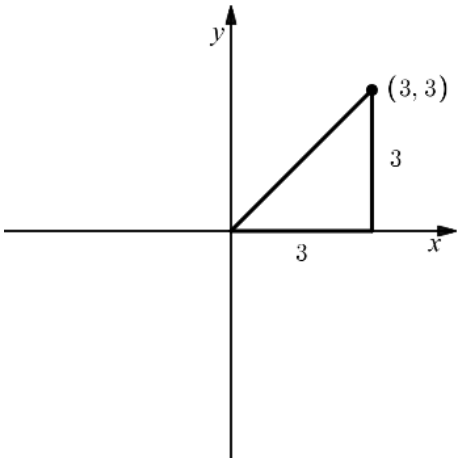
1) c)

A

2)



3) a)

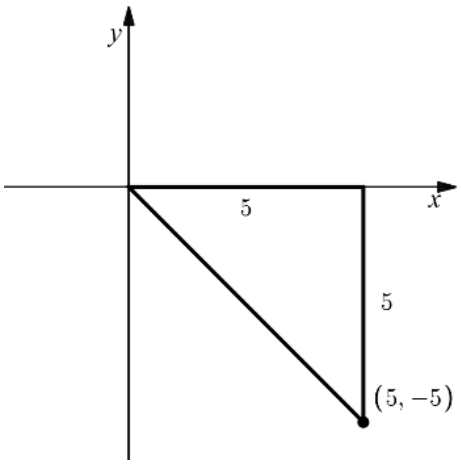


$\theta = \arctan\left(\frac{3}{3}\right) = \frac{\pi}{4}$

$r = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$

$\left(3\sqrt{2}, \frac{\pi}{4}\right)$

3) b)

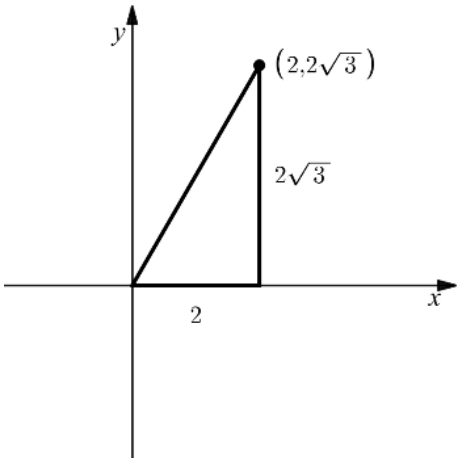


$\theta = \arctan\left(\frac{5}{5}\right) = \frac{\pi}{4}$

$r = \sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$

$\left(5\sqrt{2}, -\frac{\pi}{4}\right)$

3) c)

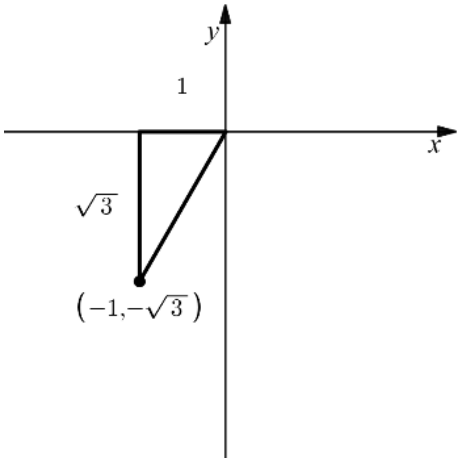


$\theta = \arctan\left(\frac{2\sqrt{3}}{2}\right) = \frac{\pi}{3}$

$r = \sqrt{2^2 + \left(2\sqrt{3}\right)^2} = \sqrt{16} = 4$

$\left(4, \frac{\pi}{3}\right)$

3) d)

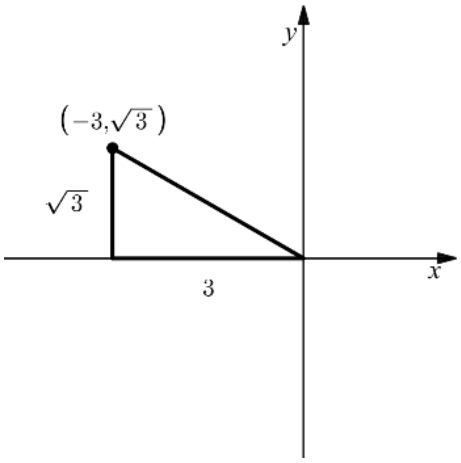


$\theta = \arctan\left(\sqrt{3}\right) = \frac{\pi}{3}$

$r = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = \sqrt{4} = 2$

$\left(2, -\frac{2\pi}{3}\right)$

3) e)

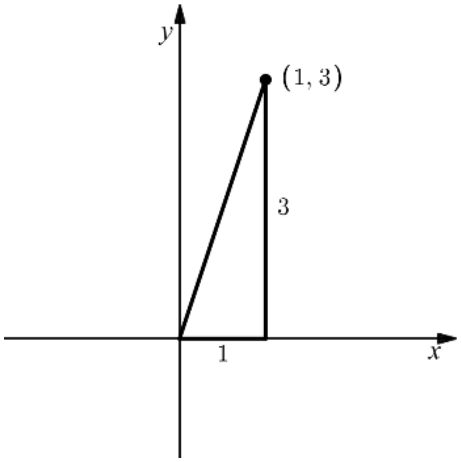


$\theta = \arctan\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$

$r = \sqrt{3^2 + \left(\sqrt{3}\right)^2} = \sqrt{12} = 2\sqrt{3}$

$\left(2\sqrt{3}, \frac{5\pi}{6}\right)$

4) a)

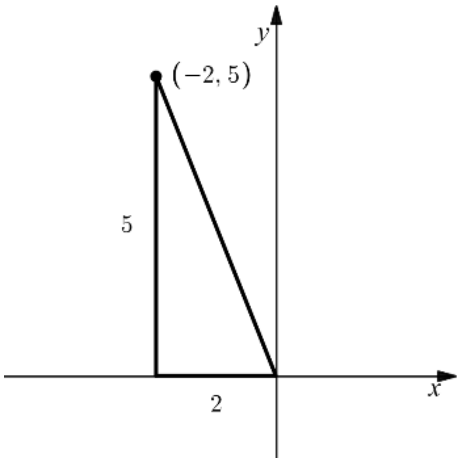


$\theta = \arctan\left(\frac{3}{1}\right) = 1.25$

$r = \sqrt{1^2 + 3^2} = \sqrt{10}$

$\left(\sqrt{10}, 1.25\right)$

4) b)

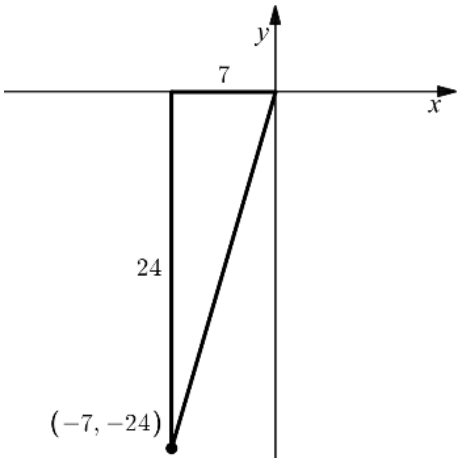


$\theta = \arctan\left(\frac{5}{2}\right) = 1.19$

$r = \sqrt{2^2 + 5^2} = \sqrt{29}$

$\left(\sqrt{29}, 1.95\right)$

4) c)

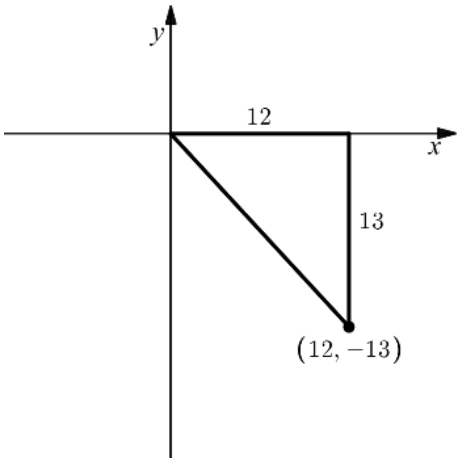


$\theta = \arctan\left(\frac{24}{7}\right) = 1.29$

$r = \sqrt{7^2 + 24^2} = \sqrt{225} = 25$

$(25, -1.85)$

4) d)



$\theta = \arctan\left(\frac{13}{12}\right) = 0.825$

$r = \sqrt{12^2 + 13^2} = \sqrt{313}$

$\left(\sqrt{313}, -0.825\right)$

5) a)

$\left(2, -\frac{3\pi}{4}\right)$

5) b)

$\left(1, -\frac{\pi}{6}\right)$

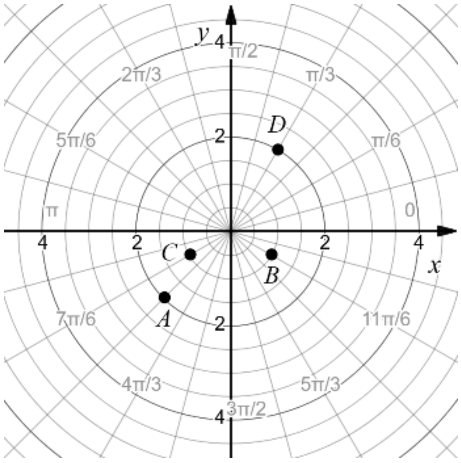
5) c)

$\left(1, -\frac{5\pi}{6}\right)$

5) d)

$\left(2, \frac{\pi}{3}\right)$

5) e)



6) a)

$\left(3\cos\frac{\pi}{4}, 3\sin\frac{\pi}{4}\right) = \left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$

6) b)

$\left(2\cos\frac{5\pi}{6}, 2\sin\frac{5\pi}{6}\right) = \left(-\sqrt{3}, 1\right)$

6) c)

$\left(-4\cos\frac{\pi}{3}, -4\sin\frac{\pi}{3}\right) = \left(-2, -2\sqrt{3}\right)$

7)

$\left(-3, \frac{\pi}{3} + \pi\right)$ so $p = \frac{4\pi}{3}$

$\left(3, \frac{\pi}{3} + 2\pi\right)$ so $q = \frac{7\pi}{3}$

8) a)

Each vertex is one rotation of $\frac{2\pi}{3}$ about the pole

$A\left(2, \frac{\pi}{6}\right)$, $B\left(2, \frac{\pi}{6} + \frac{2\pi}{3}\right)$, $C\left(2, \frac{\pi}{6} + \frac{4\pi}{3}\right)$

So

$A\left(2, \frac{\pi}{6}\right)$, $B\left(2, \frac{5\pi}{6}\right)$, $C\left(2, -\frac{\pi}{2}\right)$

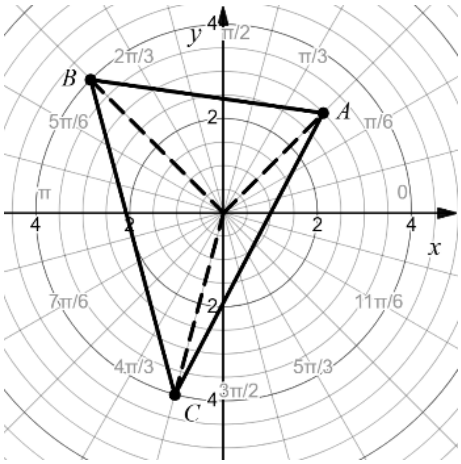
8) b)

$\frac{\pi}{3}$ is the angle to rotate from A to C about the pole

$\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$ or $\frac{\pi}{6}-\frac{\pi}{3}=-\frac{\pi}{6}$

$A\left(2,\frac{\pi}{6}\right)$, $B(0,0)$, $C\left(2,\frac{\pi}{2}\right)$ or $A\left(2,\frac{\pi}{6}\right)$, $B(0,0)$, $C\left(2,-\frac{\pi}{6}\right)$

9)



$\frac{1}{2}\times 3\times 4\times \sin\left(\frac{3\pi}{4}-\frac{\pi}{4}\right)+\frac{1}{2}\times 4\times 4\times \sin\left(\frac{17\pi}{12}-\frac{3\pi}{4}\right)+\frac{1}{2}\times 4\times 3\times \sin\left(\frac{7\pi}{12}+\frac{\pi}{4}\right)$

$=6\sin\frac{\pi}{2}+8\sin\frac{2\pi}{3}+6\sin\frac{5\pi}{6}$

$=6+4\sqrt{3}+3=9+4\sqrt{3}=15.9$ units^2

10) a)

RHS: $4-3\sin\frac{\pi}{2}=4-3=1$ so the point lies on the line.

10) b)

Since $-1\leq\sin\theta\leq 1$

$r=4-3\times(-1)=7$

So when $\theta=\frac{3\pi}{2}$

$\left(7,\frac{3\pi}{2}\right)$

10) c)

$r = 4 - 3\sin\frac{\pi}{4} = 4 - \frac{3\sqrt{2}}{2}$

$\left(4 - \frac{3\sqrt{2}}{2}\right)\cos\frac{\pi}{4} = \left(4 - \frac{3\sqrt{2}}{2}\right)\frac{\sqrt{2}}{2} = 2\sqrt{2} - \frac{3}{2}$

$\left(4 - \frac{3\sqrt{2}}{2}\right)\sin\frac{\pi}{4} = 2\sqrt{2} - \frac{3}{2}$

So $\left(2\sqrt{2} - \frac{3}{2}, 2\sqrt{2} - \frac{3}{2}\right)$

11)

The two angles must be related by:

$b + \pi = -b + 2\pi k$ for some integer k

$2b = 2\pi k - \pi$

$b = \pi k - \frac{1}{2}\pi$

$\Rightarrow b = \frac{1}{2}(2k - 1)\pi$

That is, b must be some multiple of $(2n + 1)\frac{\pi}{2}$ where $n \in \mathbb{Z}$

Or directly above or below the pole.